Alexander Martin

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EDUCATION

University of Rochester ***Rochester, New York***

Bachelor of Science in Computer Science; Honors in Research; Major GPA: 3.73 *May 2024*

RESEARCH EXPERIENCE

Formal And Computational Semantics (FACTS) Lab ***University of Rochester***

Undergraduate Researcher; advised by Dr. Aaron Steven White *May 2022 – Present*

Frames Across Multiple Sources (FAMuS):

- Created a corpus for document level information extraction and cross document argument linking.
- Designed annotator qualification tasks for document level argument annotation.
- Formulated inter-annotator agreement metrics creating customized F1 and agreement scores.
- Received the River Campus Libraries Data Set Grant for additional annotation experiments.

MegaWika (https://arxiv.org/abs/2307.07049**):**

- Created a corpus to help LLMs cite sources with report-source pairs in 50 different languages.
- Devised annotator task for verifying whether events existing in a passage existed in their citation.

Multi-Document Role Recovery:

- Created a corpus for extracting arguments across multiple documents that discuss the same event.
- Improved MegaWika web scraping pipeline for multiple source documents and incorporated new models for source validation from FAMuS for post processing.
- Designed annotation tasks for validating and correcting machine predictions for the corpus.
- Developed a computational model for consolidating event extractions from each document into a single most informative template representation.

Event Individuation through Subevent Annotation:

- Investigated the problems of individuation at the document level through subevent annotation.
- Formulated a new metric for the "range" of events unique event counts for a document.

Connotate:

- Created an open-source annotation tool for entity, event and argument, and coreference annotation.
- Generated documents for pilot tasks using event and entity trigger extraction models for FrameNet.
- Optimized pilot document quality using modified optimal matching algorithms based on gold dataset annotations from RED and machine predicted triggers for FrameNet frames.

Visual Intelligence & Social Multimedia Analytics (VIStA) Lab ***University of Rochester***

Undergraduate Researcher; advised by Dr. Jiebo Luo *August 2022 – Present*

Jurassic World Remake (https://arxiv.org/abs/2308.07316)**:**

- Created a new dataset consisting of skulls and their corresponding living animals.
- Proposed a new methodology for diffusion models to translate in good faith between domains.
- Wrote a first-author publication and delivered oral presentation to ACM Multimedia conference.

Rhythm and Emotion Aware Video Background Music Generation:

- Developed a method for contrastive video-audio pretraining, specifically for videos and music.
- Developed an unsupervised method for generating emotion and rhythm aware music for videos.

Closed Loop Corrections for Text-to-Image Generation:

- Designed a pipeline for correcting the generations of text-to-image models using a text-only LLM and visual foundation models.
- Built a pipeline for both LMMs and LLM+visual model corrections.
- Formulated reward metrics for the refinement process by weighing various scores about layout, subjects, and aesthetics.

Rochester Human Computer Interaction (ROC-HCI) Lab ***University of Rochester***
Undergraduate Researcher; advised by Dr. Ehsan Hoque *November 2022 – Present*

Hi5: Hand Pose Estimation with Zero Human Annotation:

- Created a synthetic dataset for hand pose estimation generating diverse images with game engines.
- Trained computer vision models on the dataset to prove the successes and failures.
- Received the Dean's Award in Engineering and Mathematics for my presentation of this work.

Environmental Protection Agency ***Remote***
Research Assistant; advised by Dr. Andrea Kirk and Dr. Clyde Martin *May 2022 – October 2022*

Quantifying the Impacts of PFAS Exposure (https://www.mdpi.com/1660-4601/20/5/4539)**:**

- Developed algorithms for relative importance analysis for large datasets to measure the effects of PFAS exposure on humans and their health including cancer risk and bone mineral density.

Independent Research ***Remote***
Researcher; advised by Dr. Clyde Martin

Probabilities Associated with a Dynamic Game

- Realized a close association between best rational number approximation of an irrational number and continued fraction approximations in a dynamic game using decreasing fractiosn (1/1,…,1/N)
- Utilized GPUs to perform efficient matrix computation for large probabilities up to N=10,000
- Attempted to fit continuous probability distributions to the data, but no existing distribution fit.

PAPERS, PUBLICATIONS, & TALKS

Accepted and Under Review Papers:

- **A. Martin**, H. Zheng, J. An, J. Luo "Jurassic World Remake: Bringing Ancient Fossils Back to Life via Zero-Shot Long Image-to-Image Translation" (**Accepted,** ACM MM)
- S. Vashishtha, **A. Martin**, W. Gantt, B. Van Durme, A.S. White "FAMuS: Frames Across Multiple Sources" (**Completed,** Intended TACL)
- S. Barham, O. Weller, et al. (incl. **A. Martin**) "MegaWika: Millions of reports and their sources across 50 diverse languages" (**Submitted,** ACL ARR; **Preprint,** ArXiv, Hugging Face)
- A. Kirk, A. DeStafano, **A. Martin**, K. Kirk, C. Martin "A New Interpretation of Relative Importance on An Analysis of Per and Polyfluorinated Alkyl Substances (PFAS) Exposures on Bone Mineral Density" (**Published**, IJERPH)
- **A. Martin**, C. Martin "Probabilities Associated with a Dynamic Game" (**Under Revision**, Communications in Information Systems)
- C. Martin, **A. Martin**, A. Kirk "Regression on subspaces: A tool for mixtures with applications to the effects of per-and polyfluoroalkyl substances on risk of major osteoporotic fracture" (**Under Revision**, Journal of Agricultural Biological and Environment Statistics)

In work:

- **A. Martin***, J. An*, J. Luo "Closed-Loop Corrections for Text-to-Image Generation" (Intended CVPR 2024)
- **A. Martin**, S. Vashishtha, S. Barham, B. Van Durme, A.S. White "Multi-Document Role Recovery" (Intended ACL Long 2024)
- **A. Martin**, J. Luo "Unsupervised Rhythm and Emotion Aware Background Music Generation"
- W. Gantt, **A. Martin**, A.S. White "On Event Individuation through Subevent Annotation" (Intended ACL Short 2024)
- A. Imtiaz, C. Nair, O. Hai, **A. Martin**, A.S. White "Connotate: A Concrete Annotator Tool" (Intended ACL Short 2024)
- M. Hasan, C. Ozel, **A. Martin**, S. Potter. T. Adnan, E. Hoque "Hi5: Hand Pose Estimation with Zero Human Annotation" (Intended CVPR 2024)

Invited Talks:

- ACM Multimedia 2023, *Jurassic World Remake* (October 2023)
- University of Rochester Data Set Grant Presentations: *Document Level Event Information and Relations* (September 2023)
- University of Rochester Research Symposium: *Hi5: Hand Pose Estimation with Zero Human Annotation* (April 2023)

HONORS, AWARDS, & GRANTS

River Campus Libraries Data Set Grant: *Event Keyed Summarization*. ($850)	Spring 2023
Make It Happen Grant: *Re-Vision Home Inspection*. ($500)	Spring 2022
University of Rochester Undergrad Research, *Deans' Award: Engineering and Mathematics*	2023
University of Rochester, *Residential Life Best Program of the Year*	2023

PROJECTS

ReVision Home Inspection

- Engineered a platform to automate visual home inspections to identify problems during inspections.
- Built vision models for automating home inspections and creating price estimates.
- Make it Happen Grant from the University of Rochester; Finalist Cornell Tech's Start Up Awards.

TEACHING EXPERIENCE

Computer Science Department ***Rochester, New York***

Teaching Assistant - Intro to Artificial Intelligence (CSC 242)	*Spring 2023*
Teaching Assistant - Data Structures and Algorithms (CSC 172)	*Spring 2022, Fall 2022*
Teaching Assistant - Intro to Computer Science (CSC 171)	*Fall 2021*
Computer Science Undergraduate Tutor	*Fall 2021—Present*

LEADERSHIP ACTIVITIES & VOLUNTEERING

Resident Advisor and D'Lion *August 2021 – Present*

- Fostered a healthy and inclusive environment for residents.
- Assisted first-year students with the transition to college life.

DandyHacks Finance and Logistics Coordinator *January 2021 – January 2022*

- Organized the universities largest hackathon by managing finances and donations.

Computer Science Undergraduate Council: Communications Chair *January 2021 – May 2022*

- Interacted with alumni, grad students, and faculty to organize events for computer science majors.

STEM Initiative Mentor

- Taught local students coding and hardware fundamentals for projects using Lego Robotics

SKILLS

Programming Languages: Python, Java, C++, MATLAB **Familiar:** C, SQL, JavaScript, R, Kotlin
Tools/Frameworks: PyTorch, Amazon Mechanical Turk, Docker, AWS, LaTeX, Git, Overleaf
Soft Skills: Small group instruction, conflict resolution, time management